

Principles of Qualitative Analysis

Groups I to V (only Ag^+ , Hg^{2+} , Cu^{2+} , Pb^{2+} , Fe^{3+} , Cr^{3+} , Al^{3+} , Ca^{2+} , Ba^{2+} , Zn^{2+} , Mn^{2+} and Mg^{2+}); Nitrate, halides (excluding fluoride), carbonate and bicarbonate, sulphate and sulphide.

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Groups I to V (only Ag^+ , Hg^{2+} , Cu^{2+} , Pb^{2+} , Fe^{3+} , Cr^{3+} , Al^{3+} , Ca^{2+} , Ba^{2+} , Zn^{2+} , Mn^{2+} and Mg^{2+}); Nitrate, halides (excluding fluoride), carbonate and bicarbonate, sulphate and sulphide.

Theory that governs qualitative analysis-

Two basic principles of great use in the analysis are:

- (i) the Solubility product; and
- (ii) the Common ion effect.

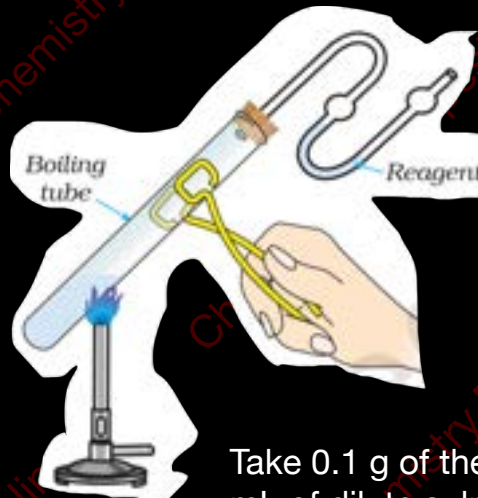
$K_{sp} \downarrow$

When ionic product of a salt exceeds its solubility product, precipitation takes place.

Ionic product of salt is controlled by making use of common ion effect

SYSTEMATIC ANALYSIS OF ANIONS

Step - I : Preliminary Test with Dilute Sulphuric Acid



Take 0.1 g of the salt in a test tube and add 1–2 mL of dilute sulphuric acid.

Observe the change, if any, at room temperature.

If no gas is evolved, warm the content of the test tube. If gas is evolved test it by using the apparatus shown

SYSTEMATIC ANALYSIS OF ANIONS

Step - I : Preliminary Test with Dilute Sulphuric Acid

Observations	Inference	
	Gas Evolved	Possible Anion
A colourless, odourless gas is evolved with brisk effervescence, which turns lime water milky.	CO_2	Carbonate (CO_3^{2-})
Colourless gas with the smell of rotten eggs is evolved which turns lead acetate paper black.	H_2S	Sulphide (S^{2-})
Colourless gas with a pungent smell, like burning sulphur which turns acidified potassium dichromate solution green.	SO_2	Sulphite (SO_3^{2-})
Brown fumes which turn acidified potassium iodide solution containing starch solution blue.	NO_2	Nitrate (NO_3^-)
Colourless vapours with smell of vinegar. Vapours turn blue litmus red.	CH_3COOH vapours	Acetate, (CH_3COO^-)

SYSTEMATIC ANALYSIS OF ANIONS

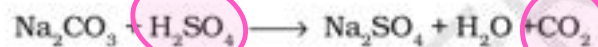
Step - I : Preliminary Test with Dilute Sulphuric Acid

NXT : Chemistry of Confirmatory Tests

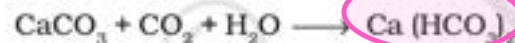
Confirmatory tests for CO_3^{2-} , S^{2-} , SO_3^{2-} , NO_2^- and CH_3COO^-

1. Test for Carbonate ion $[\text{CO}_3^{2-}]$

If there is effervescence with the evolution of a colourless and odourless gas on adding dil. H_2SO_4 to the solid salt, this indicates the presence of carbonate ion. The gas turns lime water milky due to the formation of CaCO_3 (Fig. 7.1)



If CO_2 gas is passed in excess through lime water, the milkiness produced disappears due to the formation of calcium hydrogen carbonate which is soluble in water.



SYSTEMATIC ANALYSIS OF ANIONS

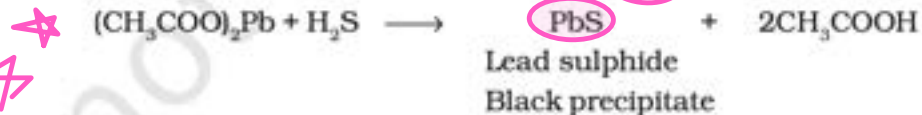
Step - I : Preliminary Test with Dilute Sulphuric Acid

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Confirmatory tests for CO_3^{2-} , S^{2-} , SO_3^{2-} , NO_2^- and CH_3COO^-

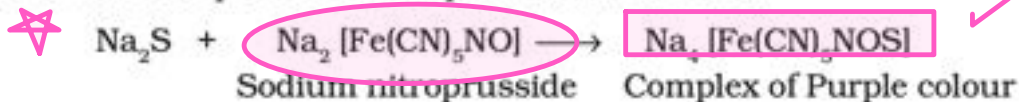
2. Test for Sulphide ion $[\text{S}^{2-}]$

- (a) With warm dilute H_2SO_4 a sulphide gives hydrogen sulphide gas which smells like rotten eggs. A piece of filter paper dipped in lead acetate solution turns black on exposure to the gas due to the formation of lead sulphide which is black in colour.



- (b) If the salt is soluble in water, take the solution of salt in water make it alkaline with ammonium hydroxide and add sodium nitroprusside solution. If it is insoluble in water take sodium carbonate extract and add a few drops of sodium nitroprusside solution. Purple or violet

colouration due to the formation of complex compound $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$ confirms the presence of sulphide ion in the salt.



SYSTEMATIC ANALYSIS OF ANIONS

Step - I : Preliminary Test with Dilute Sulphuric Acid

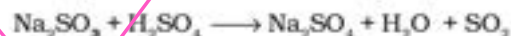
NXT : Chemistry of Confirmatory Tests

The smell of burning sulphur is caused by the release of sulphur dioxide (SO₂), a pungent, sharp, and irritating odor often compared to a freshly struck match. This toxic, colorless gas is produced when sulphur burns in the air or during industrial processes. It is distinct from, but often confused with, the "rotten egg" smell of hydrogen sulfide (H₂S)

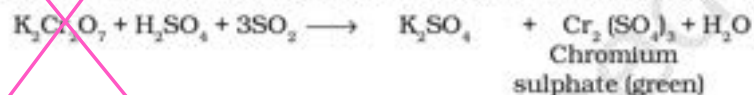
Confirmatory tests for CO₃²⁻, S²⁻, SO₃²⁻, NO₂⁻ and CH₃COO⁻

3. Test for Sulphite ion [SO₃²⁻]

- (a) On treating sulphite with warm dil. H₂SO₄, SO₂ gas is evolved which is suffocating with the smell of burning sulphur.



The gas turns potassium dichromate paper acidified with dil. H₂SO₄, green.

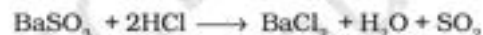


- (b) An aqueous solution or sodium carbonate extract of the salt produces a white precipitate of barium sulphite on addition of barium chloride solution.

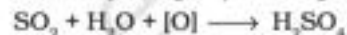
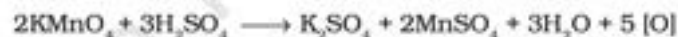


This precipitate gives following tests.

- (i) This precipitate on treatment with dilute HCl, dissolves due to decomposition of sulphite by dilute HCl. Evolved SO₂ gas can be tested.



- (ii) Precipitate of sulphite decolourises acidified potassium permanganate solution.



SYSTEMATIC ANALYSIS OF ANIONS

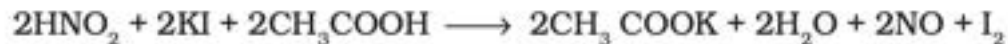
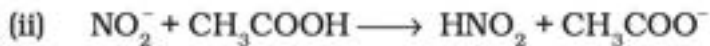
Step - I : Preliminary Test with Dilute Sulphuric Acid

NXT : Chemistry of Confirmatory Tests

Confirmatory tests for CO_3^{2-} , S^{2-} , SO_3^{2-} , NO_2^- and CH_3COO^-

4. Test for Nitrite ion $[\text{NO}_2^-]$

(a) On treating a solid nitrite with dil. H_2SO_4 and warming, reddish brown fumes of NO_2 gas are evolved. Addition of potassium iodide solution to the salt solution followed by freshly prepared starch solution and acidification with acetic acid produces blue colour. Alternatively, a filter paper moistened with potassium iodide and starch solution and a few drops of acetic acid turns blue on exposure to the gas due to the interaction of liberated iodine with starch.



SYSTEMATIC ANALYSIS OF ANIONS

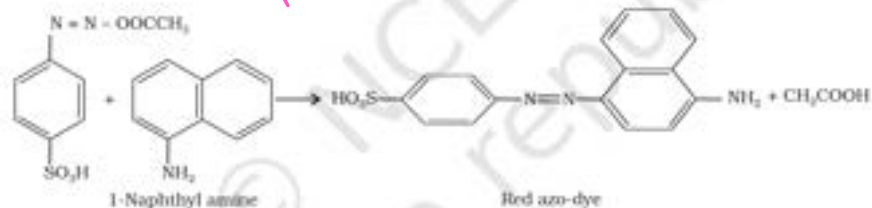
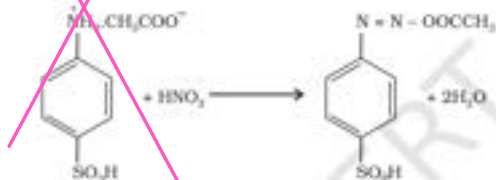
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Confirmatory tests for CO_3^{2-} , S^{2-} , SO_3^{2-} , NO_2^- and CH_3COO^-

4. Test for Nitrite ion [NO_2^-]

- (b) Sulphanilic acid — 1-naphthylamine reagent test (Griss-Ilosvay test)
On adding sulphanilic acid and 1-naphthylamine reagent to the water extract or acidified with acetic acid, sulphanilic acid is diazotised in the reaction by nitrous acid formed. Diazotised acid couples with 1-naphthylamine to form a red azo-dye.



The test solution should be very dilute. In concentrated solutions reaction does not proceed beyond diazotisation.

SYSTEMATIC ANALYSIS OF ANIONS

Step-II :If no positive result is obtained from dil. H_2SO_4 test, take 0.1 g of salt in a test tube and 3-4 drops of conc. H_2SO_4

Observations	Inference	
	Gas/Vapours Evolved	Possible Anions
A colourless gas with pungent smell, which gives dense white fumes when a rod dipped in ammonium hydroxide is brought near the mouth of the test tube.	HCl	Chloride, (Cl^-)
Reddish brown gas with a pungent odour is evolved. Intensity of reddish gas increases on heating the reaction mixture after addition of solid MnO_2 to the reaction mixture. Solution also acquires red colour.	Br_2 vapours	Bromide, (Br^-)
Violet vapours which turn starch paper blue and a layer of violet sublimate is formed on the sides of the tube. Fumes become dense on adding MnO_2 to the reaction mixture.	I_2 vapours	Iodide, (I^-)
Brown fumes evolve which become dense upon heating the reaction mixture after addition of copper turnings and the solution acquires blue colour.	NO_2	Nitrate, (NO_3^-)
Colourless, odourless gas is evolved which turns lime water milky and the gas coming out of lime water burns with a blue flame, if ignited.	CO and CO_2	Oxalate, ($\text{C}_2\text{O}_4^{2-}$)

Chemistry of Confirmatory Tests

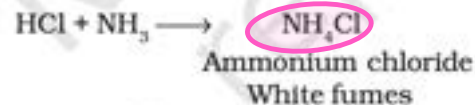
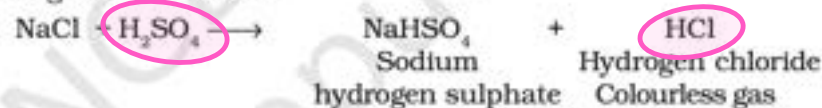
SYSTEMATIC ANALYSIS OF ANIONS

Step-II :If no positive result is obtained from dil. H₂SO₄ test, take 0.1 g of salt in a test tube and 3-4 drops of conc. H₂SO₄

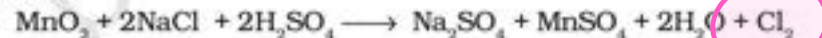
NXT: Chemistry of Confirmatory Tests

1. Test for Chloride ion [Cl⁻]

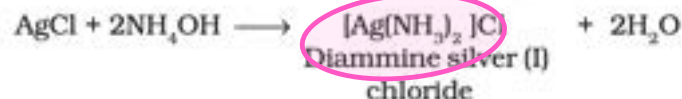
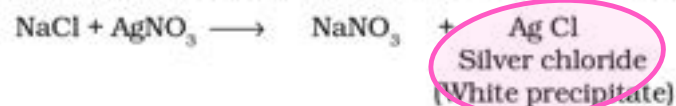
- (a) If on treatment with warm conc. H₂SO₄ the salt gives a colourless gas with pungent smell or and if the gas which gives dense white fumes with ammonia solution, then the salt may contain Cl⁻ ions and the following reaction occurs.



- (b) If a salt gives effervescence on heating with conc. H₂SO₄ and MnO₂ and a light greenish yellow pungent gas is evolved, this indicates the presence of Cl⁻ ions.



- (c) Salt solution acidified with dilute HNO₃ on addition of silver nitrate solution gives a curdy white precipitate soluble in ammonium hydroxide solution. This indicates the presence of Cl⁻ ions in the salt.



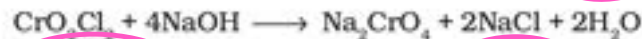
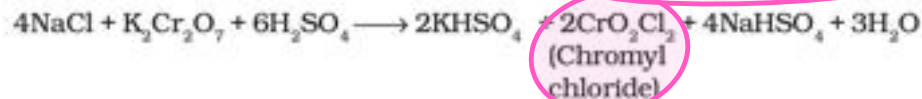
Chemistry of Confirmatory Tests

SYSTEMATIC ANALYSIS OF ANIONS

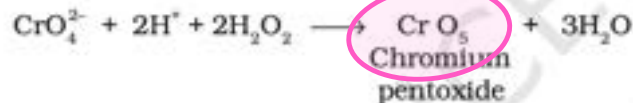
Step-II :If no positive result is obtained from dil. H₂SO₄ test, take 0.1 g of salt in a test tube and 3-4 drops of conc. H₂SO₄

NXT: Chemistry of Confirmatory Tests

- (d) Mix a little amount of salt and an equal amount of solid potassium dichromate (K₂Cr₂O₇) in a test tube and add conc. H₂SO₄ to it. Heat the test tube and pass the evolved gas through sodium hydroxide solution. If a yellow solution is obtained, divide the solution into two parts. Acidify the first part with acetic acid and then add lead acetate solution. Formation of a yellow precipitate of lead chromate confirms the presence of chloride ions in the salt. This test is called **chromyl chloride test**.



Acidify the second part with dilute sulphuric acid and add small amounts of amyl alcohol and then 1 mL of 10% hydrogen peroxide solution. On gentle shaking organic layer turns blue. CrO₄²⁻ ion formed in the reaction of chromyl chloride with sodium hydroxide reacts with hydrogen peroxide to form chromium pentoxide (CrO₅) (See structure) which dissolves in amyl alcohol to give blue colour.



SYSTEMATIC ANALYSIS OF ANIONS

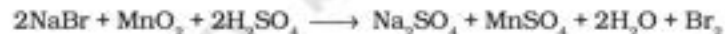
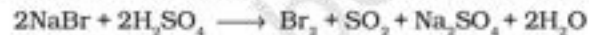
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NXT: Chemistry of Confirmatory Tests

Chemistry of Confirmatory Tests

2. Test for Bromide ion (Br⁻)

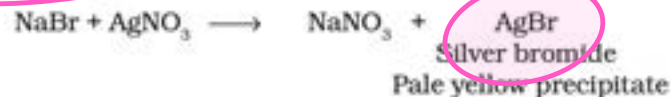
If on heating the salt with conc. H₂SO₄, reddish brown fumes of bromine are evolved in excess, this indicates the presence of Br⁻ ions. The fumes get intensified on addition of MnO₂. Bromine vapours turn starch paper yellow.



- (a) Add 1 mL of carbon tetrachloride (CCl₄)/chloroform (CHCl₃)** and excess of freshly prepared chlorine water dropwise to the salt solution in water or sodium carbonate extract neutralised with dilute HCl. Shake the test tube vigorously. The appearance of an orange brown colouration in the organic layer due to the dissolution of bromine in it, confirms the presence of bromide ions.



- (b) Acidify the sodium carbonate extract of the salt with dil. HNO₃. Add silver nitrate (AgNO₃) solution and shake the test tube. A pale yellow precipitate is obtained which dissolves in ammonium hydroxide with difficulty.



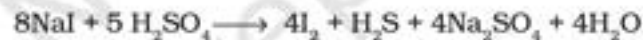
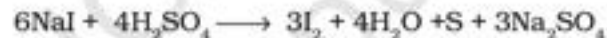
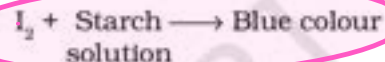
SYSTEMATIC ANALYSIS OF ANIONS

Step-II :If no positive result is obtained from dil. H₂SO₄ test, take 0.1 g of salt in a test tube and 3-4 drops of conc. H₂SO₄

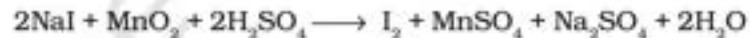
NXT: Chemistry of Confirmatory Tests

3. Test for Iodide ion (I⁻)

- (a) If on heating the salt with conc. H₂SO₄, deep violet vapours with a pungent smell are evolved. These turns starch paper blue and a violet sublimate is formed on the sides of the test tube, it indicates the presence of I⁻ ions. Some HI, sulphur dioxide, hydrogen sulphide, and sulphur are also formed due to the following reactions.



On adding MnO₂ to the reaction mixture, the violet vapours become dense.



Iodine,
harmful by
inhalation
and contact
with skin



Chlorine,
toxic by
inhalation



Chloroform,
harmful and
toxic by
inhalation



Chemistry of Confirmatory Tests

SYSTEMATIC ANALYSIS OF ANIONS

Step-II :If no positive result is obtained from dil. H₂SO₄ test, take 0.1 g of salt in a test tube and 3-4 drops of conc. H₂SO₄

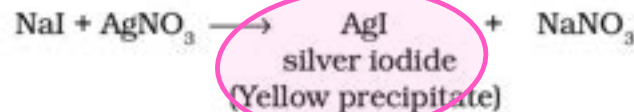
NXT: Chemistry of Confirmatory Tests

- (b) Add 1 mL of CHCl₃ or CCl₄ and chlorine water in excess to the salt solution in water or sodium carbonate extract neutralised with dil. HCl and shake the test tube vigorously. Presence of violet colouration in the organic layer confirms the presence of iodide ions.



Iodine dissolves in the organic solvent and the solution becomes violet.

- (c) Acidify sodium carbonate extract of the salt with dil. HNO₃ and add AgNO₃ solution. Appearance of a yellow precipitate insoluble in excess of NH₄OH confirms the presence of iodide ions.



Chemistry of Confirmatory Tests

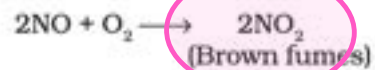
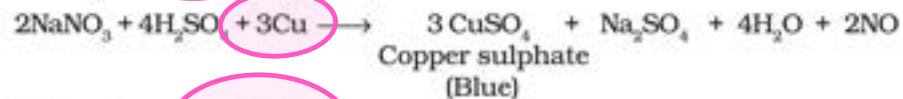
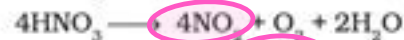
SYSTEMATIC ANALYSIS OF ANIONS

Step-II :If no positive result is obtained from dil. H₂SO₄ test, take 0.1 g of salt in a test tube and 3-4 drops of conc. H₂SO₄

NXT: Chemistry of Confirmatory Tests

4. Test for Nitrate ion [NO₃]

- (a) If on heating the salt with conc. H₂SO₄ light brown fumes are evolved then heat a small quantity of the given salt with few copper turnings or chips and conc. H₂SO₄. Evolution of excess of brown fumes indicates the presence of nitrate ions. The solution turns blue due to the formation of copper sulphate.



Chemistry of Confirmatory Tests

SYSTEMATIC ANALYSIS OF ANIONS

Step-II :If no positive result is obtained from dil. H₂SO₄ test, take 0.1 g of salt in a test tube and 3-4 drops of conc. H₂SO₄

NXT: Chemistry of Confirmatory Tests

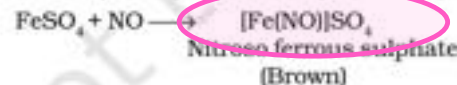
Fe⁺

4. Test for Nitrate ion [NO₃]

- (b) Take 1 mL of an aqueous solution of the salt and add 2 mL conc. H₂SO₄ slowly. Mix the solutions thoroughly and cool the test tube under the tap. Now, add freshly prepared ferrous sulphate solution along the sides of the test tube dropwise so that it forms a layer on the top of the liquid already present in the test tube. A dark brown ring is formed at the junction of the two solutions due to the formation of nitroso ferrous sulphate (Fig. 7.2). Alternatively first ferrous sulphate is added and then concentrated sulphuric acid is added.



Fig. 7.2 : Formation of brown ring



SYSTEMATIC ANALYSIS OF ANIONS

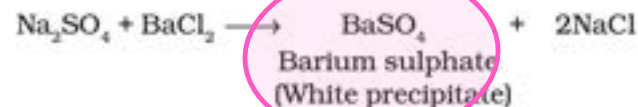
Step-III : Test for Sulphate and Phosphate

If no positive test is obtained in Steps-I and II, then tests for the presence of sulphate and phosphate ions are performed

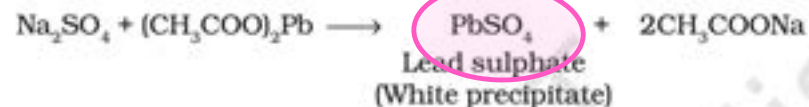
Step-III : Test for Sulphate and Phosphate

1. Test of Sulphate ions [SO_4^{2-}]

- (a) Aqueous solution or sodium carbonate extract of the salt acidified with acetic acid on addition of barium chloride gives a white precipitate of barium sulphate insoluble in conc. HCl or conc. HNO_3 .



- (b) Sulphate ions give white precipitate of lead sulphate when aqueous solution or sodium carbonate extract neutralised with acetic acid is treated with lead acetate solution.



SYSTEMATIC ANALYSIS OF ANIONS

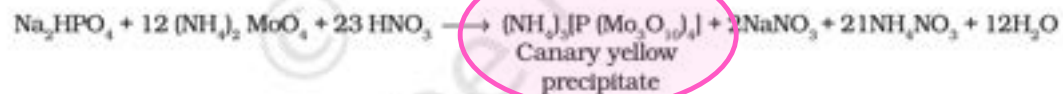
Step-III : Test for Sulphate and Phosphate

If no positive test is obtained in Steps-I and II, then tests for the presence of sulphate and phosphate ions are performed

Step-III : Test for Sulphate and Phosphate

2. Test for Phosphate ion $[\text{PO}_4^{3-}]$

- (a) Add conc. HNO_3 and ammonium molybdate solution to the test solution containing phosphate ions and boil. A yellow colouration in solution or a canary yellow precipitate of ammonium-phosphomolybdate, $(\text{NH}_4)_3[\text{P}(\text{Mo}_3\text{O}_{10})_4]$ is formed. Each oxygen of phosphate has been replaced by Mo_3O_{10} group.



+5 & +6

Step - I : Preliminary Examination of the Salt for Identification of Cation

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

1. Colour Test

Colour	Cations Indicated
Light green, Yellow, Brown	Fe^{2+} , Fe^{3+}
Blue	Cu^{2+}
Bright green	Ni^{2+}
Blue, Red, Violet, Pink	Co^{2+}
Light pink	Mn^{2+}

Ferrous - Green: Ferrous oxide is black, but hydrated ferrous salts (like ferrous sulfate) and aqueous solutions are typically light green.

Ferric - Red/Brown: Ferric oxide is a reddish-brown powder (commonly known as rust or hematite)

Ferrosoferric oxide (Fe_3O_4), a combination of both, is black.

Step - I : Preliminary Examination of the Salt for Identification of Cation

2. Dry Heating Test

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

Colour when cold	Colour when hot	Inference
Blue	White	Cu^{2+}
Green	Dirty white or yellow	Fe^{2+}
White	Yellow	Zn^{2+}
Pink	Blue	Co^{2+}

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

Step - I : Preliminary Examination of the Salt for Identification of Cation

3. Flame Test

Dip the clean loop of the platinum wire in this paste and introduce the loop in the non-luminous (oxidising) flame.

Observe the colour of the flame first with the naked eye and then through a blue glass and identify the metal ion with the help of

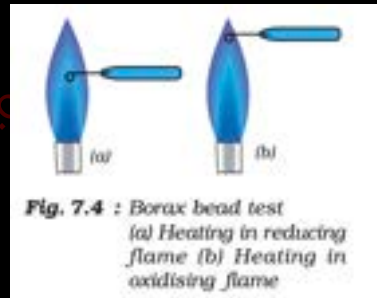
Colour of the flame observed by naked eye	Colour of the flame observed through blue glass	Inference
Green flame with blue centre	Same colour as observed without glass	Cu^{2+}
Crimson red	Purple	Sr^{2+}
Apple green	Bluish green	Ba^{2+}
Brick red	Green	Ca^{2+}

Step - I : Preliminary Examination of the Salt for Identification of Cation

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

4. Borax Bead Test



Heating in oxidising (non-luminous) flame		Heating in reducing (luminous) flame		Inference
Colour of the salt bead		Colour of the salt bead		
In cold	In hot	In cold	In hot	
Blue	Green	Red opaque	Colourless	Cu^{2+}
Reddish brown	Violet	Grey	Grey	Ni^{2+}
Light violet	Light violet	Colourless	Colourless	Mn^{2+}
Yellow	Yellowish brown	Green	Green	Fe^{3+}

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

Step - I : Preliminary Examination of the Salt for Identification of Cation

5. Charcoal Cavity Test

Observations	Inference
Yellow residue when hot and grey metal when cold	Pb^{2+}
White residue with the odour of garlic	As^{3+}
Brown residue	Cd^{2+}
Yellow residue when hot and white when cold	Zn^{2+}

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

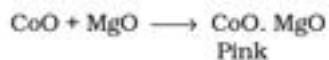
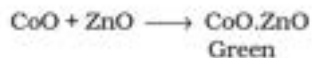
Step - I : Preliminary Examination of the Salt for Identification of Cation

6. Cobalt Nitrate Test

If the residue in the charcoal cavity is white, cobalt nitrate test is performed.

On heating, cobalt nitrate decomposes into cobalt (II) oxide, which gives a characteristic colour with metal oxide present in the cavity.

Thus, with ZnO , Al_2O_3 and MgO , the following reactions occur.



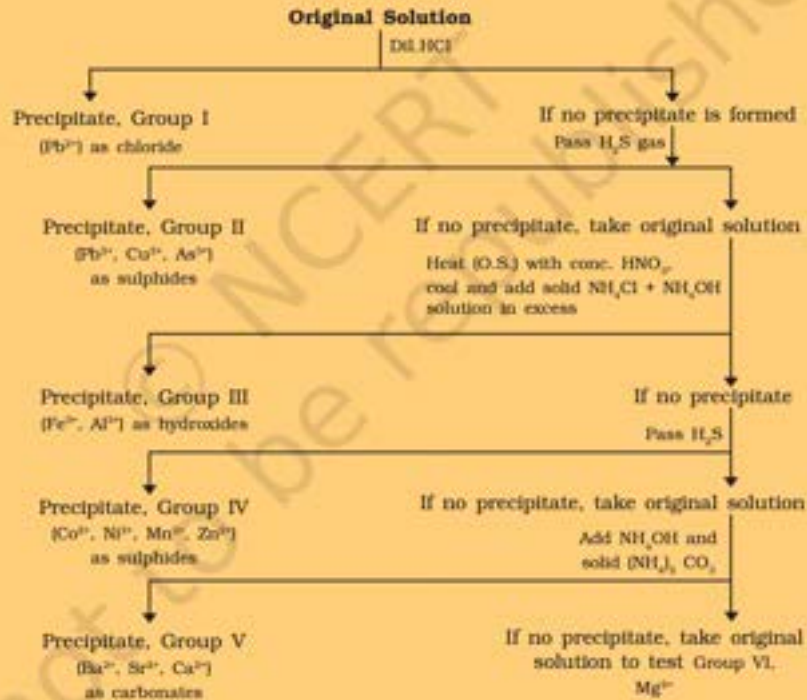
Group Analysis

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Flow Chart*



* This flow chart is for the detection of one cation only. For detection of more than one cation modification will be required.

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Group Analysis

Table 7.11 : Group reagents for precipitating ions

Group	Cations*	Group Reagent
Group zero	NH_4^+	None
Group-I	Pb^{2+}	Dilute HCl
Group-II	$\text{Pb}^{2+}, \text{Cu}^{2+}, \text{As}^{3+}$	H_2S gas in presence of dil. HCl
Group-III	$\text{Al}^{3+}, \text{Fe}^{3+}$	NH_4OH in presence of NH_4Cl
Group-IV	$\text{Co}^{2+}, \text{Ni}^{2+}, \text{Mn}^{2+}, \text{Zn}^{2+}$	H_2S in presence of NH_4OH
Group-V	$\text{Ba}^{2+}, \text{Sr}^{2+}, \text{Ca}^{2+}$	$(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4OH
Group-VI	Mg^{2+}	None

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Group Analysis

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Group zero	NH_4^+	None
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Group-II	Pb^{2+} , Cu^{2+} , As^{3+}	H_2S gas in presence of dil. HCl
Group-III	Al^{3+} , Fe^{3+}	NH_4OH in presence of NH_4Cl
Group-IV	Co^{2+} , Ni^{2+} , Mn^{2+} , Zn^{2+}	H_2S in presence of NH_4OH
Group-V	Ba^{2+} , Sr^{2+} , Ca^{2+}	$(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4OH
Group-VI	Mg^{2+}	None

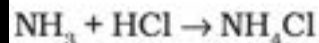
SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

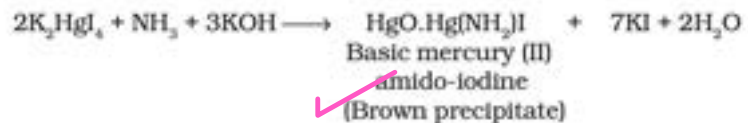
NXT>>>>

Group Analysis

Analysis of Zero group cation (NH_4^+ ion)



On passing the gas through Nessler's reagent, a brown colouration or a precipitate of basic mercury(II) amido-iodine is formed.



Step - I : Preliminary Examination of the Salt for Identification of Cation

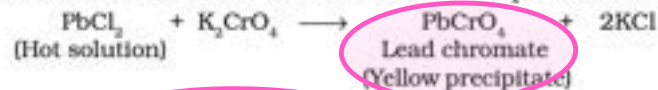
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Analysis of Group-I cations

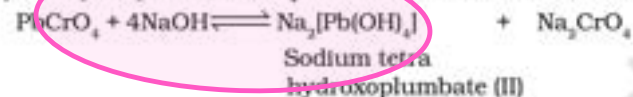
- $$\text{PbCl}_2 + 2\text{KI} \longrightarrow \text{PbI}_2 + 2\text{KCl}$$
- (Hot solution) Yellow precipitate

This yellow precipitate (PbI_2) is soluble in boiling water and reappears on cooling as shining crystals.

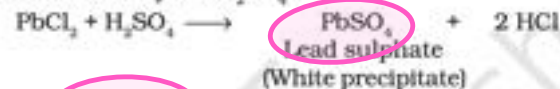
2. On addition of potassium chromate (K_2CrO_4) solution a yellow precipitate of lead chromate is obtained. This confirms the presence of Pb^{2+} ions.



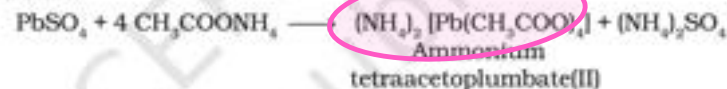
The yellow precipitate (PbCrO_4) is soluble in hot NaOH solution.



3. A white precipitate of lead sulphate (PbSO_4) is formed on addition of alcohol followed by dil. H_2SO_4 .



Lead sulphate is soluble in ammonium acetate solution due to the formation of tetraacetoplumbate(II) ions. This reaction may be promoted by addition of few drops of acetic acid.



SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

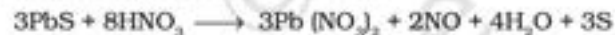
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Group Analysis

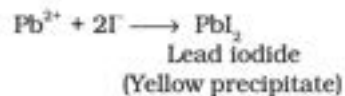
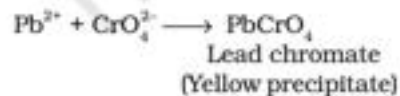
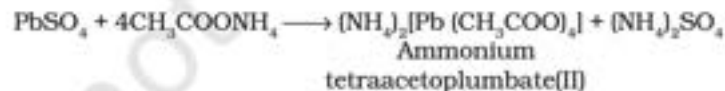
Analysis of Group-II cations

1. Test for Lead ion (Pb^{2+})

Lead sulphide precipitate dissolves in dilute HNO_3 . On adding dil. H_2SO_4 and a few drops of alcohol to this solution a white precipitate of lead sulphate appears. This indicates the presence of lead ions.



The white precipitate dissolves in ammonium acetate solution on boiling. When this solution is acidified with acetic acid and potassium chromate solution is added, a yellow precipitate of PbCrO_4 is formed. On adding potassium iodide solution, a yellow precipitate of lead iodide is formed.



SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

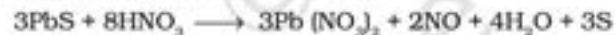
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Group Analysis

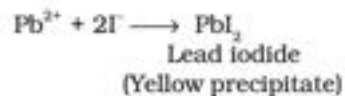
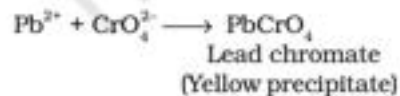
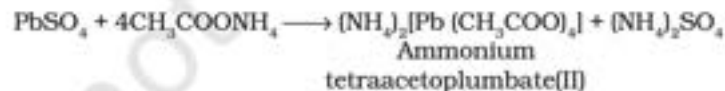
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SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Group Analysis

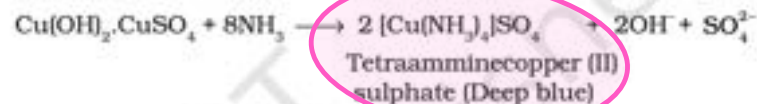
Analysis of Group-II cations

2. Test for Copper ion (Cu^{2+})

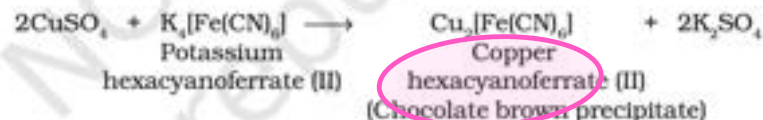
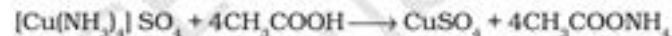
- (a) Copper sulphide dissolves in nitric acid due to the formation of copper nitrate.



On heating the reaction mixture for long time, sulphur is oxidised to sulphate and copper sulphate is formed and the solution turns blue. A small amount of NH_4OH precipitates basic copper sulphate which is soluble in excess of ammonium hydroxide due to the formation of tetraamminecopper (II) complex.



- (b) The blue solution on acidification with acetic acid and then adding potassium ferrocyanide $[\text{K}_4\text{Fe}(\text{CN})_6]$ solution gives a chocolate colouration due to the formation of copper ferrocyanide i.e. $\text{Cu}_2[\text{Fe}(\text{CN})_6]$.



SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

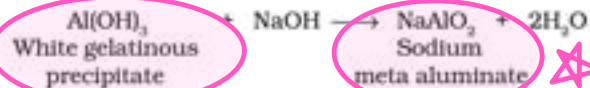
Group Analysis

Analysis of Group-III cations

1. Test for Aluminium ions (Al^{3+})

- (a) When the solution containing aluminium chloride is treated with sodium hydroxide, a white gelatinous precipitate of aluminium hydroxide is formed

which is soluble in excess of sodium hydroxide solution due to the formation of sodium meta aluminate.



- (b) In the second test when blue litmus is added to the solution, a red colouration is obtained due to the acidic nature of the solution. On addition of NH_4OH solution drop by drop the solution becomes alkaline and aluminium hydroxide is precipitated. Aluminium hydroxide adsorbs blue colour from the solution and forms insoluble adsorption complex named 'lake'. Thus a blue mass floating in the colourless solution is obtained. The test is therefore called **lake test**.

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Group Analysis

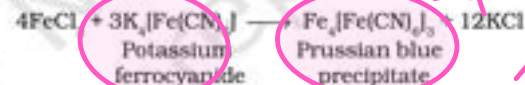
Analysis of Group-III cations

2. Test for ferric ions (Fe^{3+})

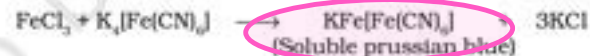
Reddish brown precipitate of ferric hydroxide dissolves in hydrochloric acid and ferric chloride is formed.



- (a) When the solution containing ferric chloride is treated with potassium ferrocyanide solution a blue precipitate/colouration is obtained. The colour of the precipitate is Prussian blue. It is ferric ferrocyanide. The reaction takes place as follows:



If potassium hexacyanoferrate (III) (i.e. potassium ferrocyanide) is added in excess then a product of composition $\text{KFe}[\text{Fe}(\text{CN})_6]$ is formed. This tends to form a colloidal solution ('soluble Prussian blue') and cannot be filtered.



- (b) To the second part of the solution, add potassium thiocyanate (potassium sulphocyanide) solution. The appearance of a blood red colouration confirms the presence of Fe^{3+} ions.



Group Analysis

Analysis of Group-III cations

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Procedure: To the original solution (O.S.), add a pinch of solid NH_4Cl , boil it, and then add NH_4OH solution in excess.

Observation: A green gelatinous precipitate indicates the presence of Cr^{3+} .

A. Sodium Hydroxide and Bromine Water Test

1. Dissolve the green precipitate in a small amount of dilute HCl .
2. Add excess $NaOH$ and a few drops of bromine water (or H_2O_2).
3. Heat the mixture.

• **Observation:** The solution turns yellow due to the formation of Sodium Chromate (Na_2CrO_4).

• **Equation:**



Group Analysis

Analysis of Group-III cations

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

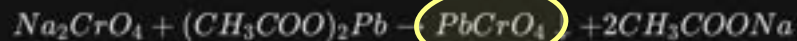
NXT>>>>>

Observation: A green gelatinous precipitate indicates the presence of Cr^{3+} .

B. Lead Acetate Test

Take the yellow solution obtained from the previous test, acidify it with acetic acid, and add Lead Acetate solution.

- **Observation:** A bright yellow precipitate of Lead Chromate ($PbCrO_4$) confirms Chromium.
- **Equation:**



C. Silver Nitrate Test

Acidify the yellow chromate solution with acetic acid and add Silver Nitrate ($AgNO_3$).

- **Observation:** A brick-red precipitate of Silver Chromate (Ag_2CrO_4) is formed.

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Group Analysis

Analysis of Group-III cations

Observation: A green gelatinous precipitate indicates the presence of Cr^{3+} .

3. The "Chromyl Chloride" Test (For Chloride & Chromium Confirmation)

While usually used to detect Chlorides, this can also confirm the presence of Chromium in a solid salt.

1. Mix the salt with solid $K_2Cr_2O_7$ and concentrated H_2SO_4 .
2. Heat the mixture.

- **Observation:** Reddish-brown vapors of Chromyl Chloride (CrO_2Cl_2) are evolved. Passing these vapors into $NaOH$ turns the solution yellow.

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Group Analysis

Analysis of group-IV cations

1. Test for Zinc ion (Zn^{2+})

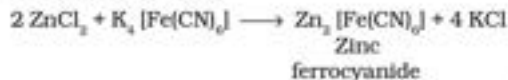
Zinc sulphide dissolves in hydrochloric acid to form zinc chloride.



- (a) On addition of sodium hydroxide solution it gives a white precipitate of zinc hydroxide, which is soluble in excess of NaOH solution on heating. This confirms the presence of Zn^{2+} ions.



- (b) When potassium ferrocyanide $\text{K}_4\text{Fe(CN)}_6$ solution is added to the solution after neutralisation by NH_4OH solution, a white or a bluish white precipitate of zinc ferrocyanide appears.



Group Analysis

Analysis of group-IV cations

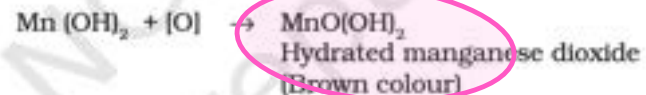
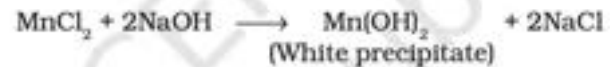
SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

2. Test for Manganese ion (Mn^{2+})

Manganese sulphide precipitate dissolves in dil. HCl on boiling. On addition of NaOH solution in excess, a white precipitate of manganese hydroxide is formed which turns brown due to atmospheric oxidation into hydrated manganese dioxide.



SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

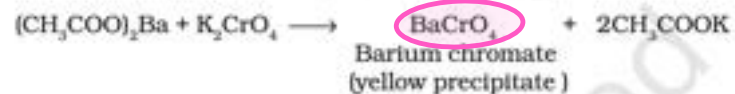
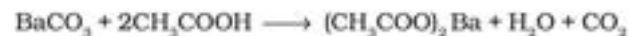
NXT>>>>>

Group Analysis

Analysis of Group-V cations

1. Test for Barium ion (Ba^{2+})

- (a) Potassium chromate (K_2CrO_4) solution gives a yellow precipitate of barium chromate when the solution of fifth group precipitate in acetic acid is treated with it.



- (b) **Flame test :** Take a platinum wire and dip it in conc. HCl. Heat it strongly until the wire does not impart any colour to the non-luminous flame. Now dip the wire in the paste of the (Group-V) precipitate in conc. HCl. Heat it in the flame. A grassy green colour of the flame confirms the presence of Ba^{2+} ions.

Group Analysis

Analysis of Group-V cations

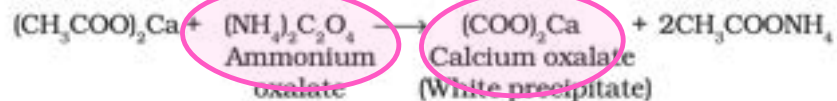
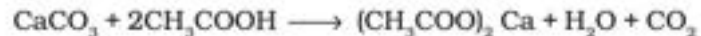
SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

3. Test for Calcium ion (Ca^{2+})

- (a) Solution of the fifth group precipitate in acetic acid gives a white precipitate with ammonium oxalate solution.



- (b) **Flame test :** Perform the flame test as mentioned above. Calcium imparts brick red colour to the flame which looks greenish-yellow through blue glass.

SYSTEMATIC ANALYSIS OF CATIONS

Step - I : Preliminary Examination of the Salt for Identification of Cation

NXT>>>>>

Group Analysis

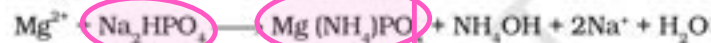
Analysis of Group-VI cations

Test for Magnesium ion (Mg^{2+})

- (a) If group-V is absent the solution may contain magnesium carbonate, which is soluble in water in the presence of ammonium salts because the equilibrium is shifted towards the right hand side.



The concentration of carbonate ions required to produce a precipitate is not attained. When disodium hydrogenphosphate solution is added and the inner walls of the test tube are scratched with a glass rod, a white crystalline precipitate of magnesium ammonium phosphate is formed which indicates the presence of Mg^{2+} ions.

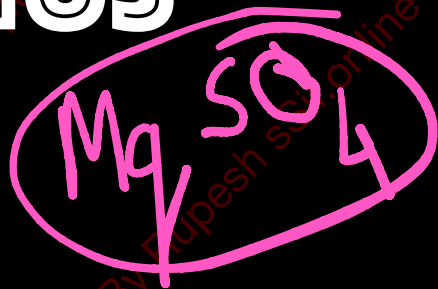


Magnesium ammonium
phosphate (White precipitate)

BONUS

Sl. No.	Experiment	Observation	Inference
1.	Noted the colour of the given salt.	White	Cu^{2+} , Fe^{2+} , Ni^{2+} , Co^{2+} , Mn^{2+} are absent.
2.	Noted the smell of the salt.	No specific smell.	S^{2-} , SO_3^{2-} , CH_3COO^- may be absent.
3.	Heated 0.5 g of the salt in a dry test tube and noted the colour of the gas evolved and change in the colour of the residue on heating and cooling.	(i) No gas was evolved. (ii) No particular change in colour of the residue is observed when heated and when cooled.	(i) CO_3^{2-} may be present, NO_3^- , NO_2^- , Br^- may be absent. (ii) Zn^{2+} may be absent.
4.	Prepared a paste of the salt with conc. HCl and performed the flame test.	No distinct colour of the flame seen.	Ca^{2+} , Sr^{2+} , Ba^{2+} , Cu^{2+} may be absent.
5.	Borax bead test was not performed as the salt was white in colour.	—	—
6.	Treated 0.1 g of salt with 1 mL dil. H_2SO_4 and warmed.	No effervescence and evolution of vapours.	CO_3^{2-} , SO_3^{2-} , S^{2-} , NO_3^- , CH_3COO^- absent.
7.	Heated 0.1 g of salt with 1 mL conc. H_2SO_4 .	No gas evolved.	Cl^- , Br^- , I^- , NO_3^- , $\text{C}_2\text{O}_4^{2-}$ are absent.
8.	Acidified 1 mL of aqueous salt solution with conc. HNO_3 . Warmed the contents and then added 4-5 drops of ammonium molybdate solution.	No yellow precipitate	PO_4^{3-} absent.

BONUS



9.	Acidified water extract of the salt with dil. HCl and then added 2mL of $BaCl_2$ solution.	A white ppt. is obtained which is insoluble in conc. HNO_3 and conc. HCl.	SO_4^{2-} present.
10.	Heated 0.1 g of salt with 2 mL NaOH solution.	Ammonia gas is not evolved.	NH_4^+ absent.
11.	Attempted to prepare original solution of the salt by dissolving 1g of it in 20 mL water.	Clear solution formed	Water soluble salt is present.
12.	To a small part of the above salt solution added 2 mL of dil. HCl.	No white precipitate formed.	Group-I absent.
13.	Passed H_2S gas through one portion of the solution of step 12.	No precipitate formed.	Group-II absent.
14.	Since salt is white, heating with conc. HNO_3 is not required. Added about 0.2 g of solid ammonium chloride and then added excess of ammonium hydroxide to the solution of step 12.	No precipitate formed.	Group-III absent.
15.	Passed H_2S gas through the above solution.	No precipitate formed.	Group-IV absent.
16.	Added excess of ammonium hydroxide solution to the original solution and then added 0.5 g of ammonium carbonate.	No precipitate formed.	Group-V absent.
17.	To the original solution of salt added ammonium hydroxide solution, followed by disodium hydrogen phosphate solution. Heated and scratched the sides of the test tube.	White precipitate.	Mg^{2+} confirmed.